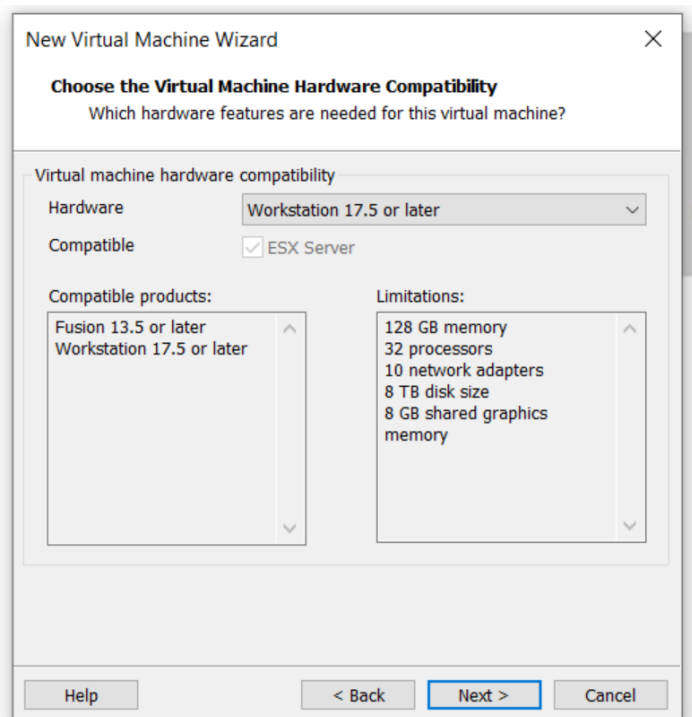
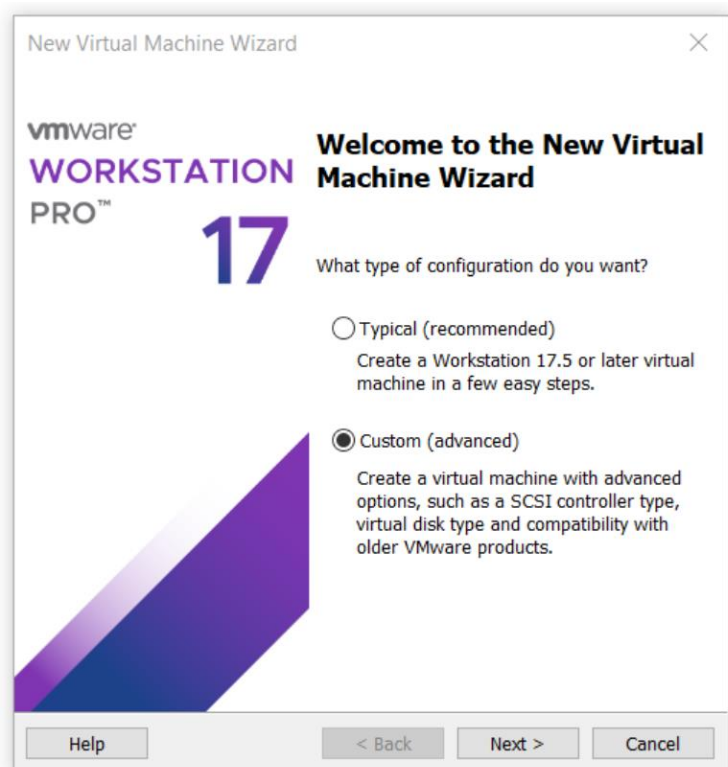


TP Serveur DHCP & Serveur DNS :

1) Création de la machine SRVUBU25



New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☐ Installer disc image file (iso):

C:\Users\ivann\Downloads\ubuntu-16.04.7-desktop-æ

Browse...

☒ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Select a Guest Operating System

Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help

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Next >

Cancel

New Virtual Machine Wizard

✕

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

SRVUBU25

Location:

C:\Users\ivann\Documents\Virtual Machines\SRVUBU25

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

✕

Processor Configuration
Specify the number of processors for this virtual machine.

Processors

Number of processors:

2

Number of cores per processor:

1

Total processor cores:

2

Help

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Next >

Cancel

New Virtual Machine Wizard

Memory for the Virtual Machine

How much memory would you like to use for this virtual machine?

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Memory for this virtual machine:

4096

MB

Maximum recommended memory:

13.2 GB

Recommended memory:

4 GB

Guest OS recommended minimum:

2 GB

Help

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Cancel

New Virtual Machine Wizard

Network Type

What type of network do you want to add?

Network connection

☐ Use bridged networking

Give the guest operating system direct access to an external Ethernet network. The guest must have its own IP address on the external network.

☒ Use network address translation (NAT)

Give the guest operating system access to the host computer's dial-up or external Ethernet network connection using the host's IP address.

☐ Use host-only networking

Connect the guest operating system to a private virtual network on the host computer.

☐ Do not use a network connection

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

×

Select I/O Controller Types

Which SCSI controller type would you like to use for SCSI virtual disks?

I/O controller types

SCSI Controller:

☐ BusLogic (Not available for 64-bit guests)

☒ LSI Logic (Recommended)

☐ LSI Logic SAS

☐ Paravirtualized SCSI

Help

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Cancel

New Virtual Machine Wizard

×

Select a Disk Type

What kind of disk do you want to create?

Virtual disk type

☐ IDE

☒ SCSI (Recommended)

☐ SATA

☐ NVMe

Help

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Cancel

New Virtual Machine Wizard

Select a Disk

Which disk do you want to use?

Disk

☒ Create a new virtual disk

A virtual disk is composed of one or more files on the host file system, which will appear as a single hard disk to the guest operating system. Virtual disks can easily be copied or moved on the same host or between hosts.

☐ Use an existing virtual disk

Choose this option to reuse a previously configured disk.

☐ Use a physical disk (for advanced users)

Choose this option to give the virtual machine direct access to a local hard disk. Requires administrator privileges.

Help

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Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☐ Allocate all disk space now.

Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

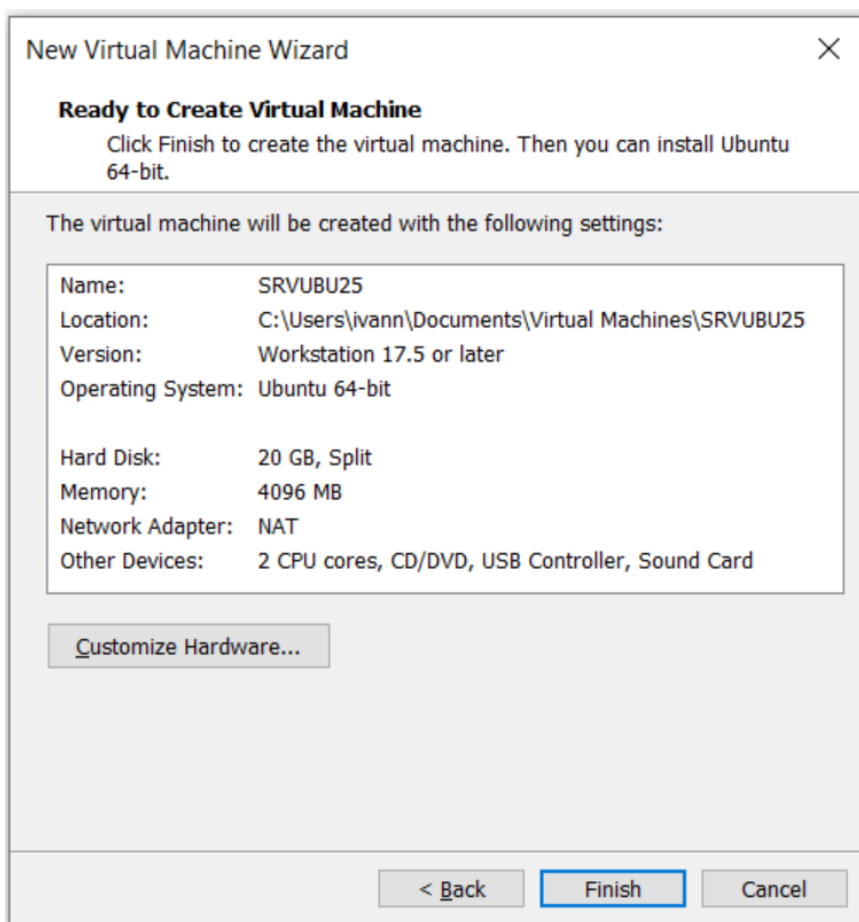
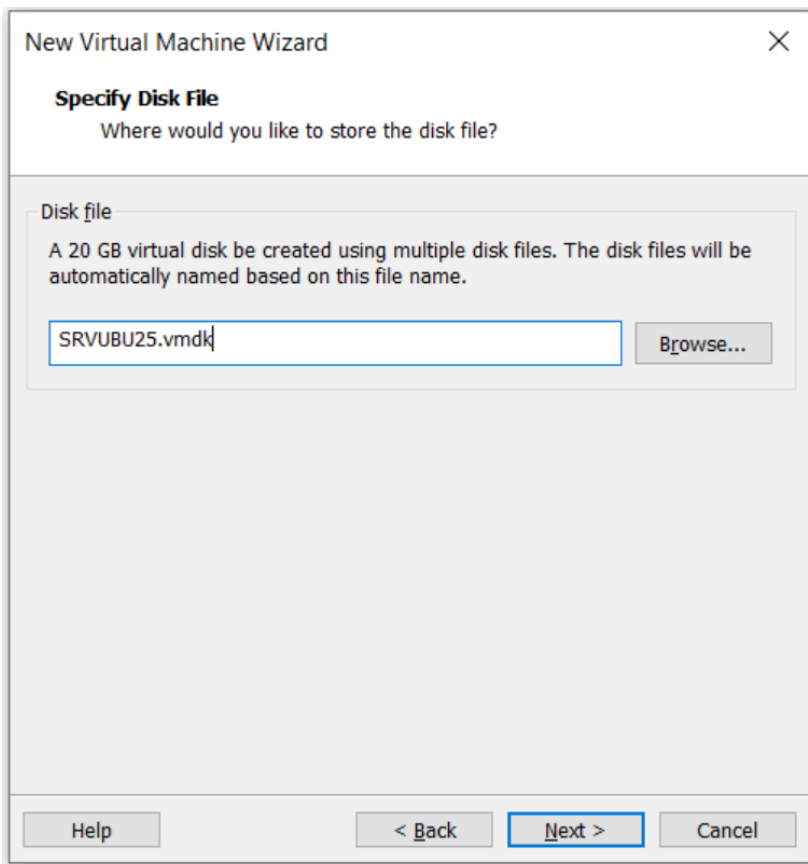
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

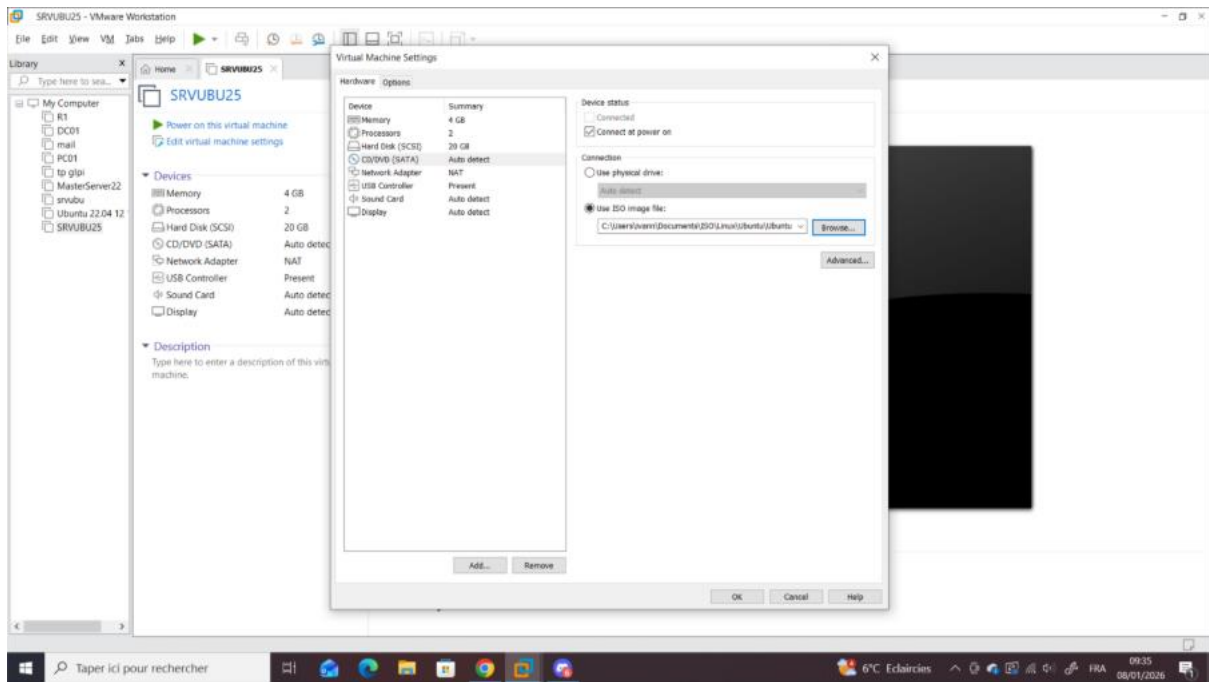
Help

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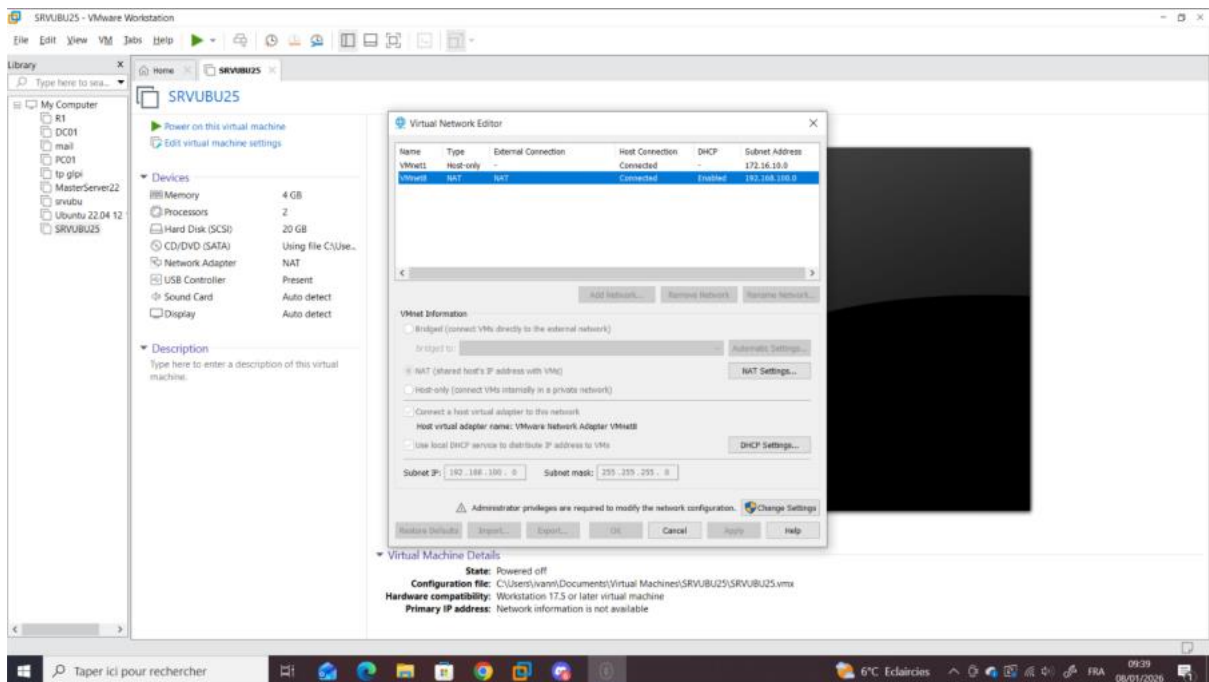
Next >

Cancel





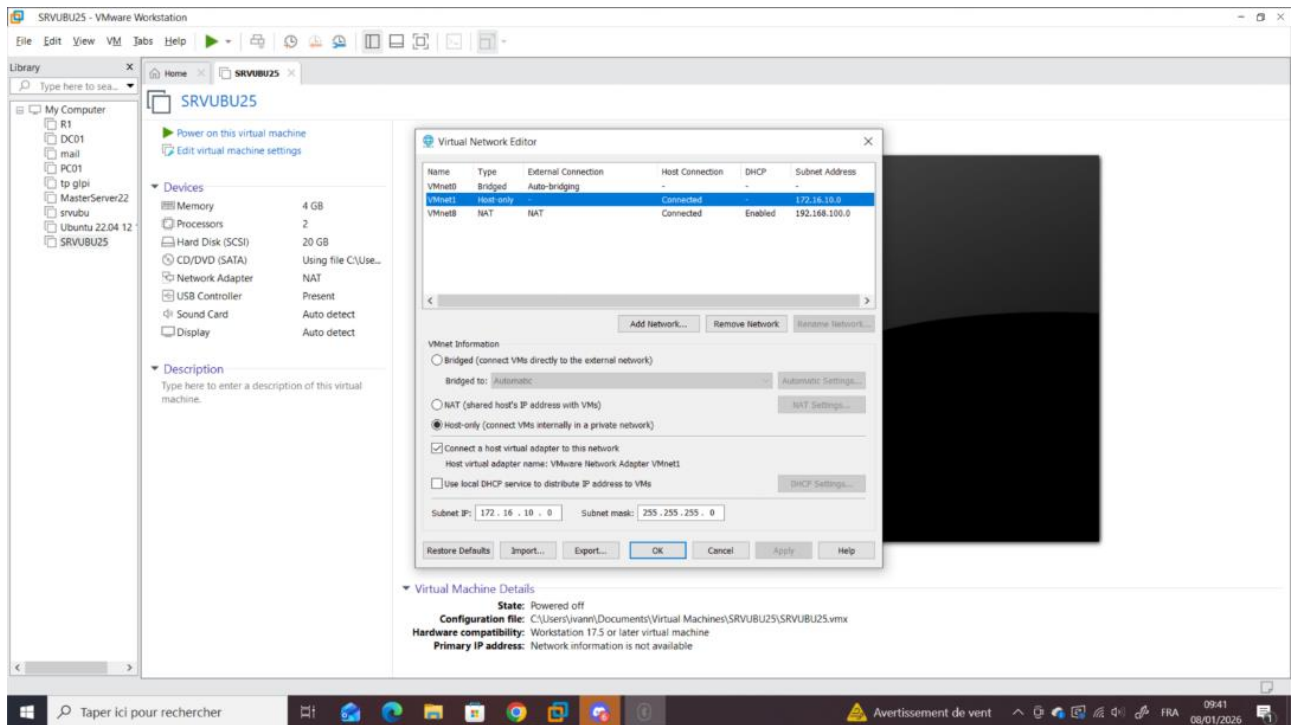
2) configuration des switch virtuel et des adresses réseau



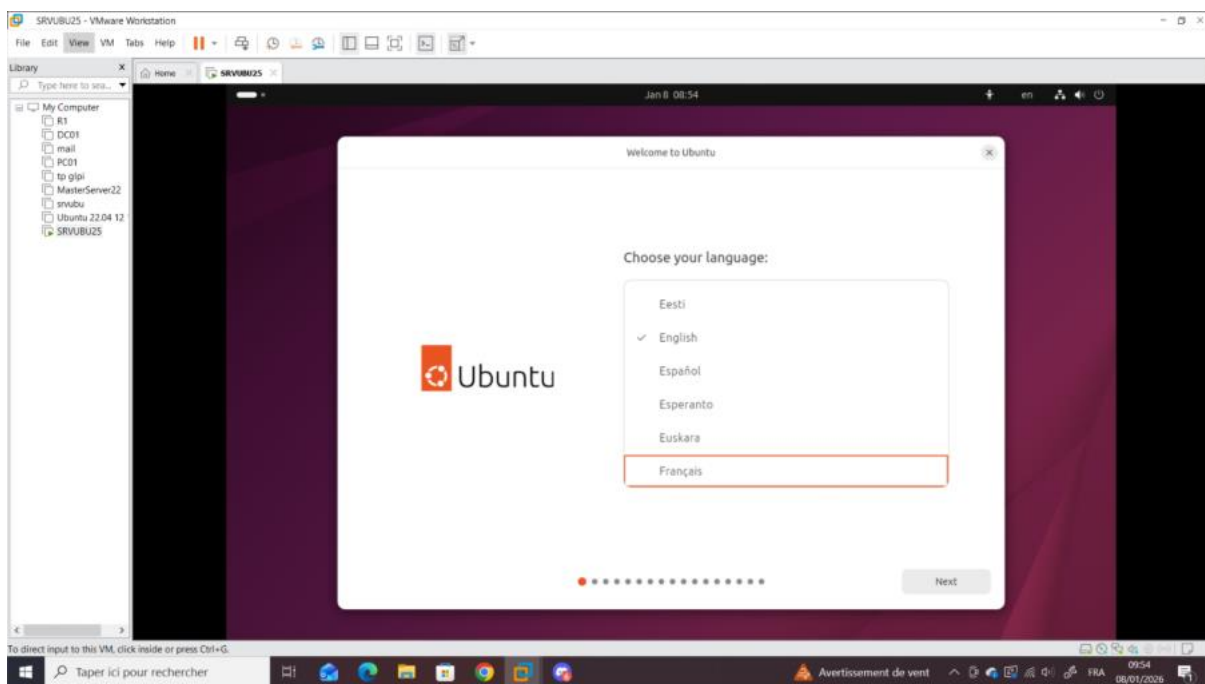
Mise en place des adresses :

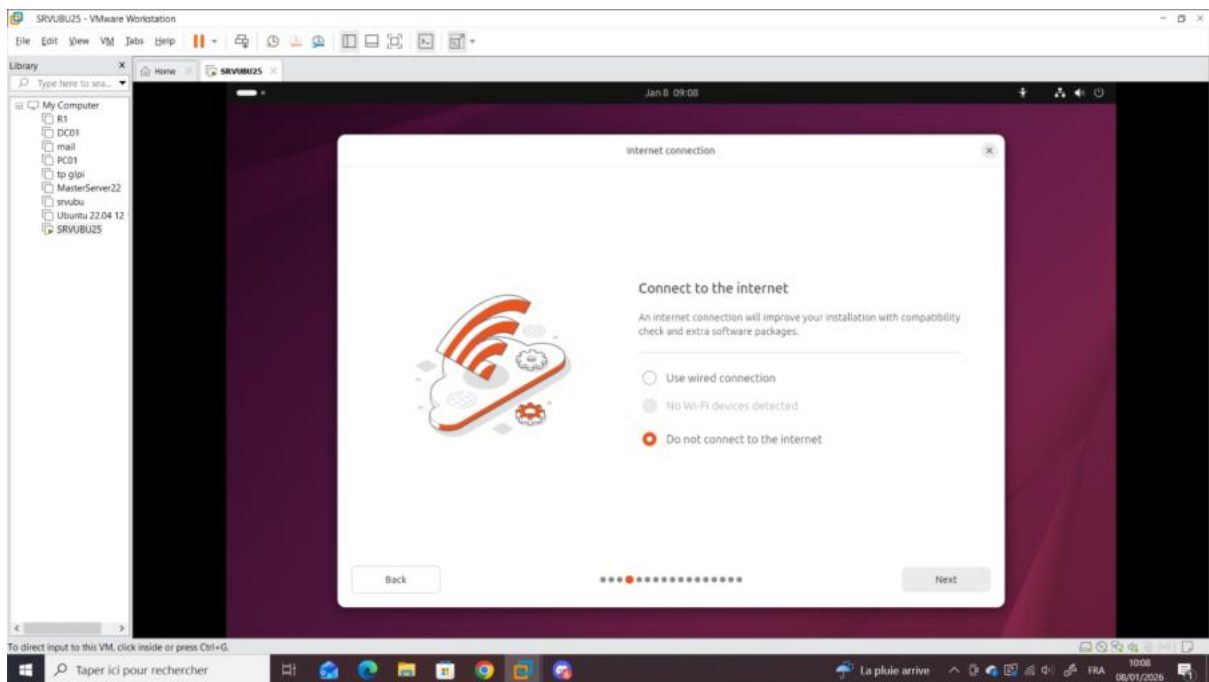
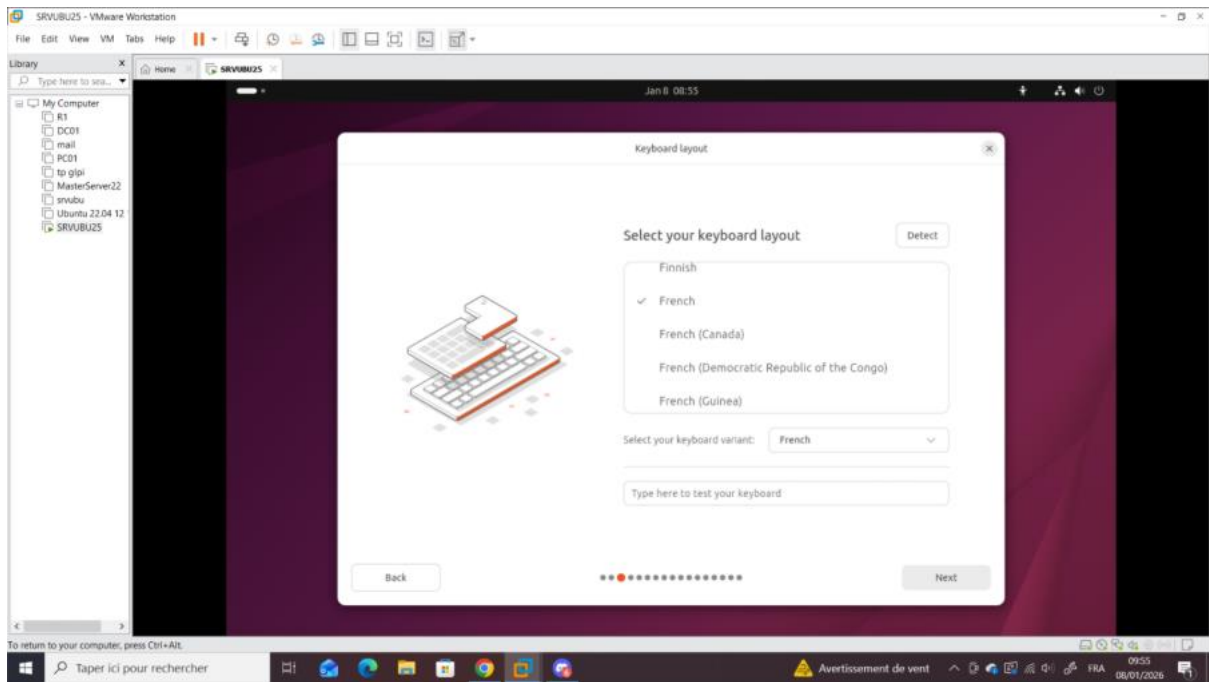
VMnet1 : 172.16.10.0/24

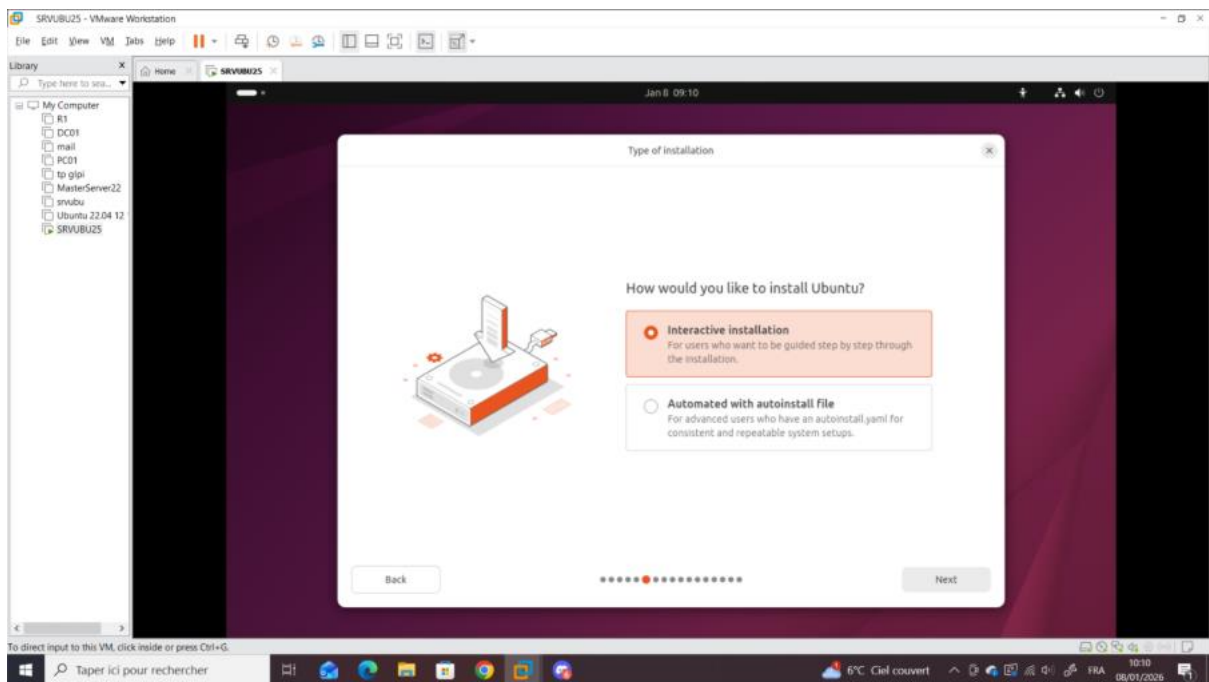
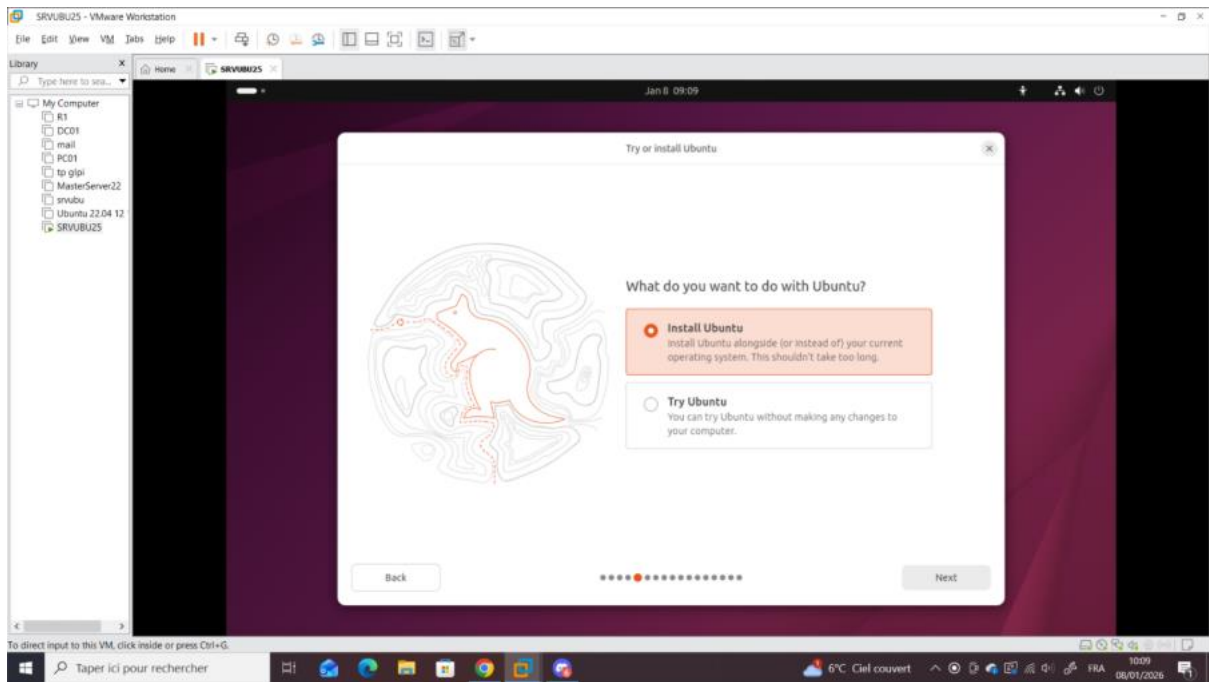
VMnet 8 : 192.168.100.0

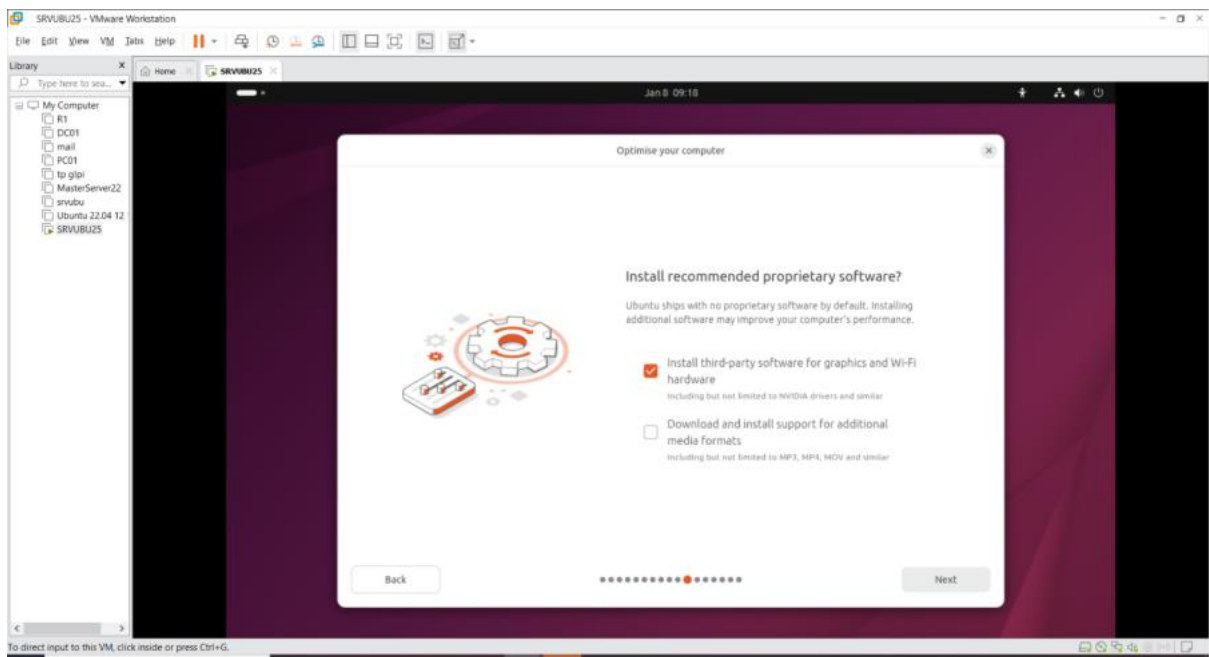
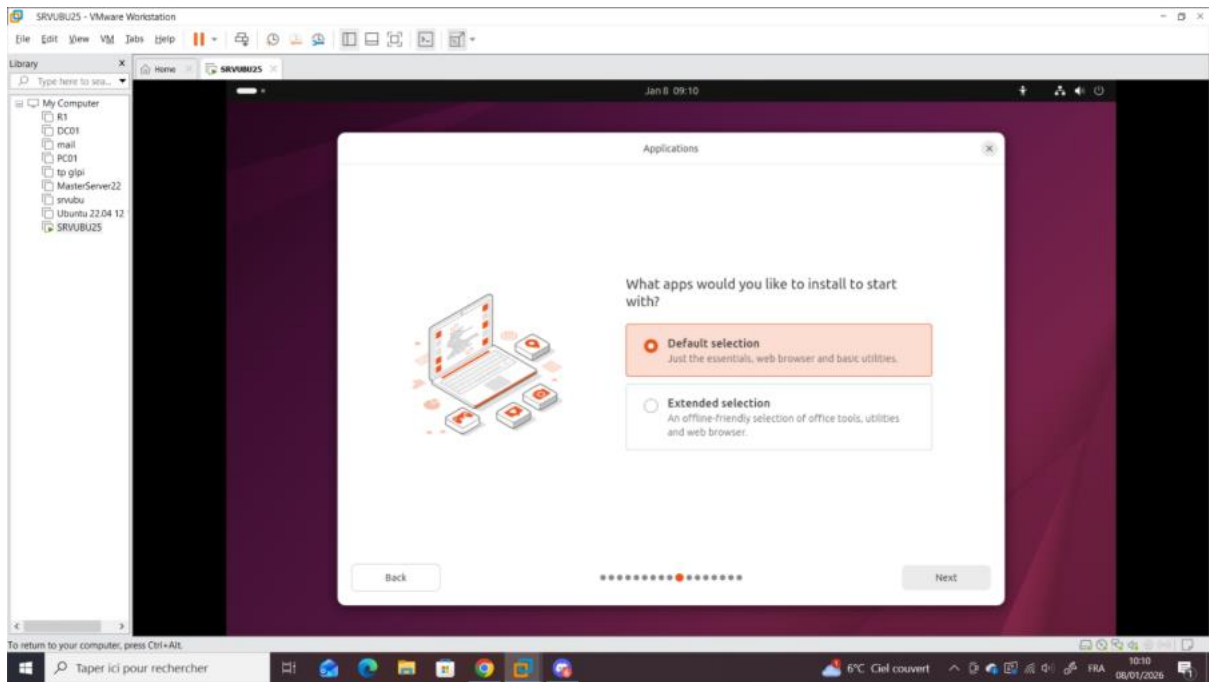


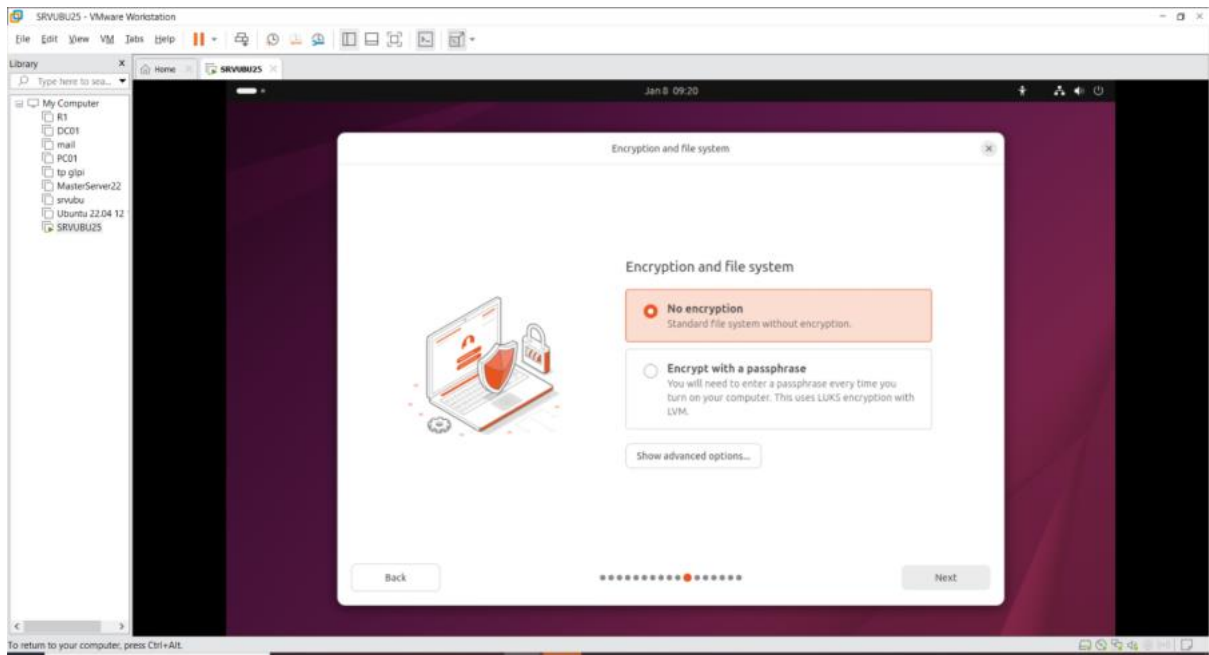
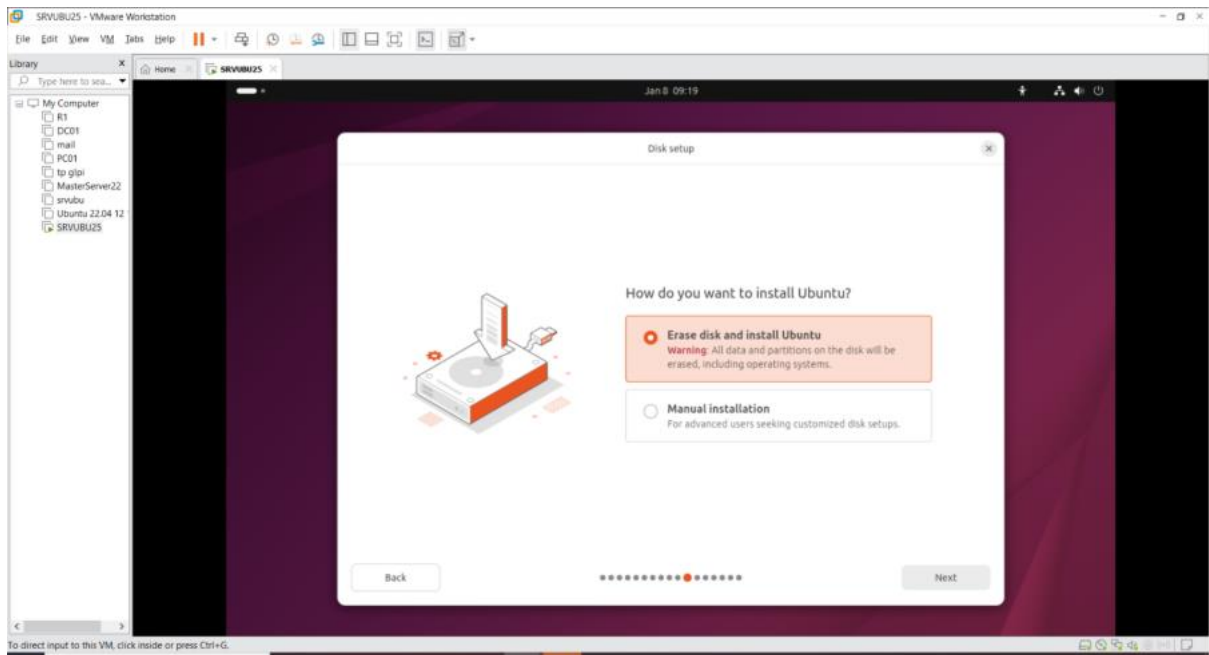
3) Installation et configuration de la machine SRVUBU25

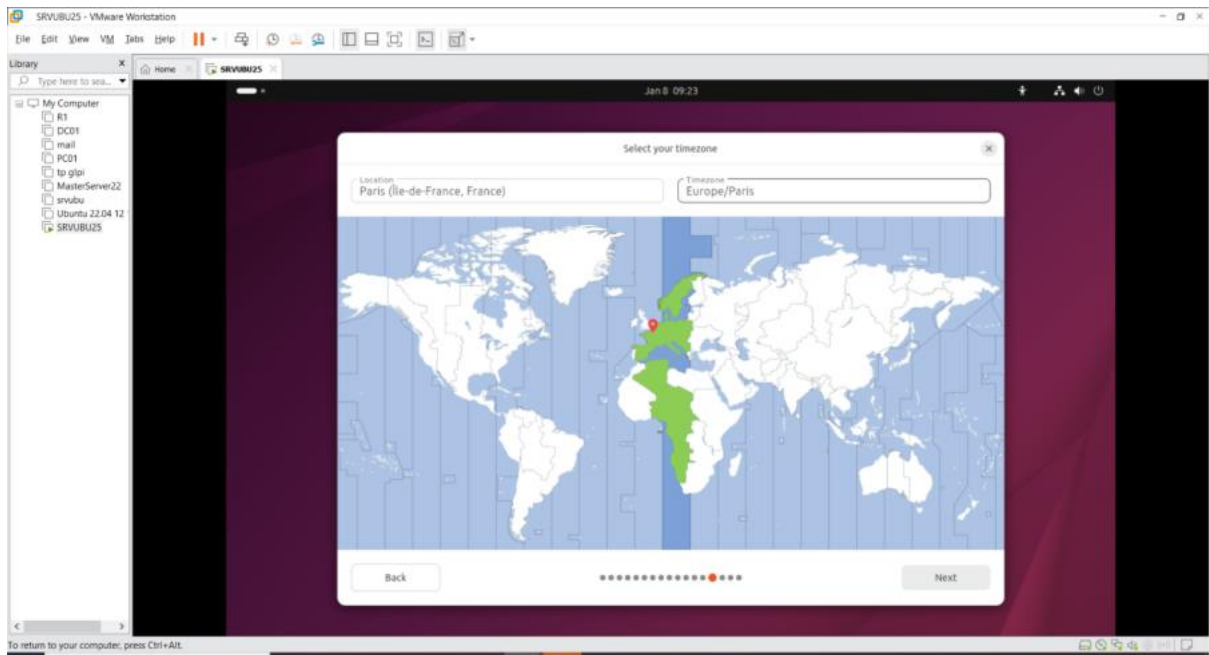
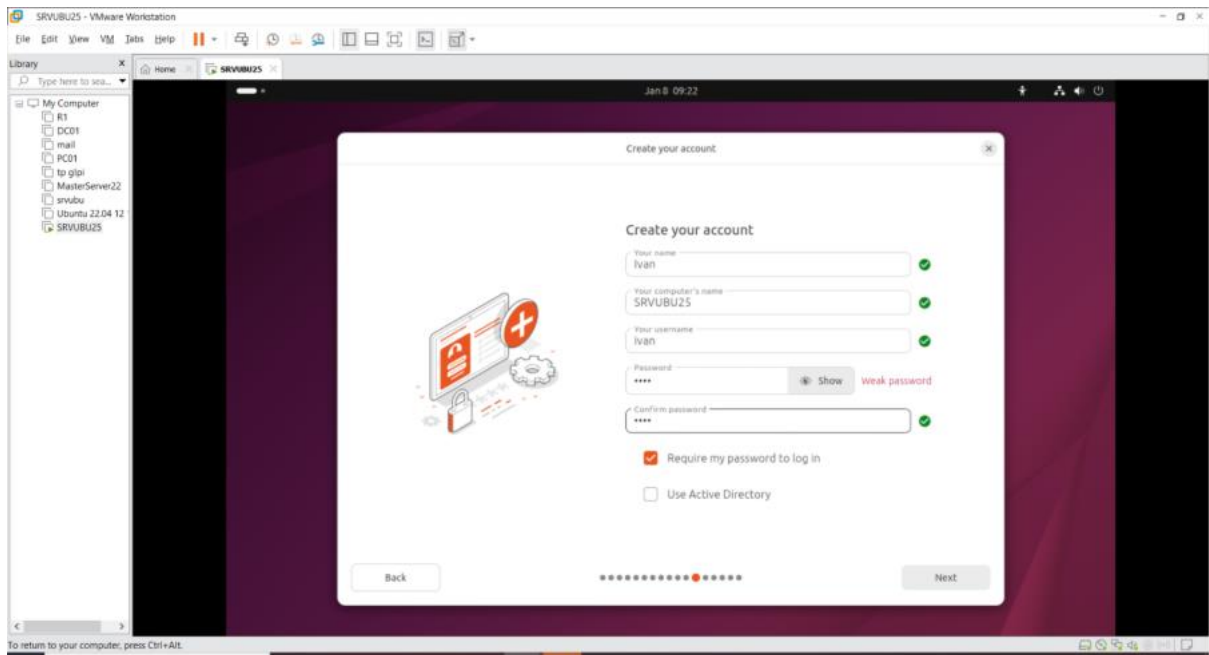


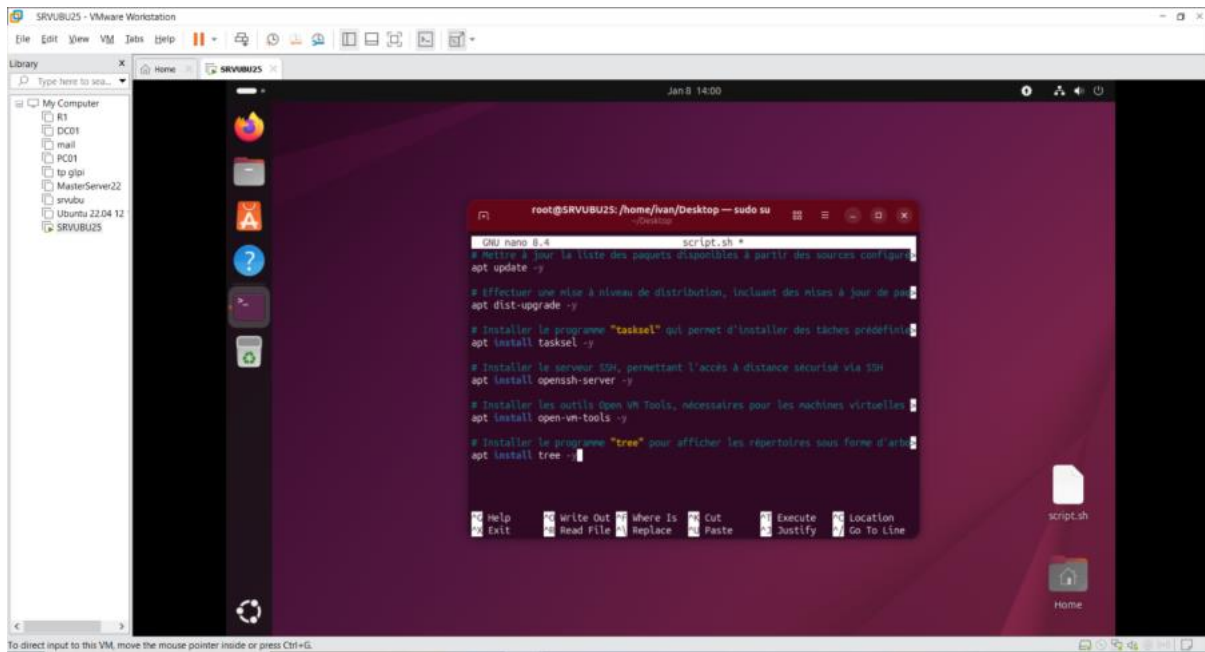
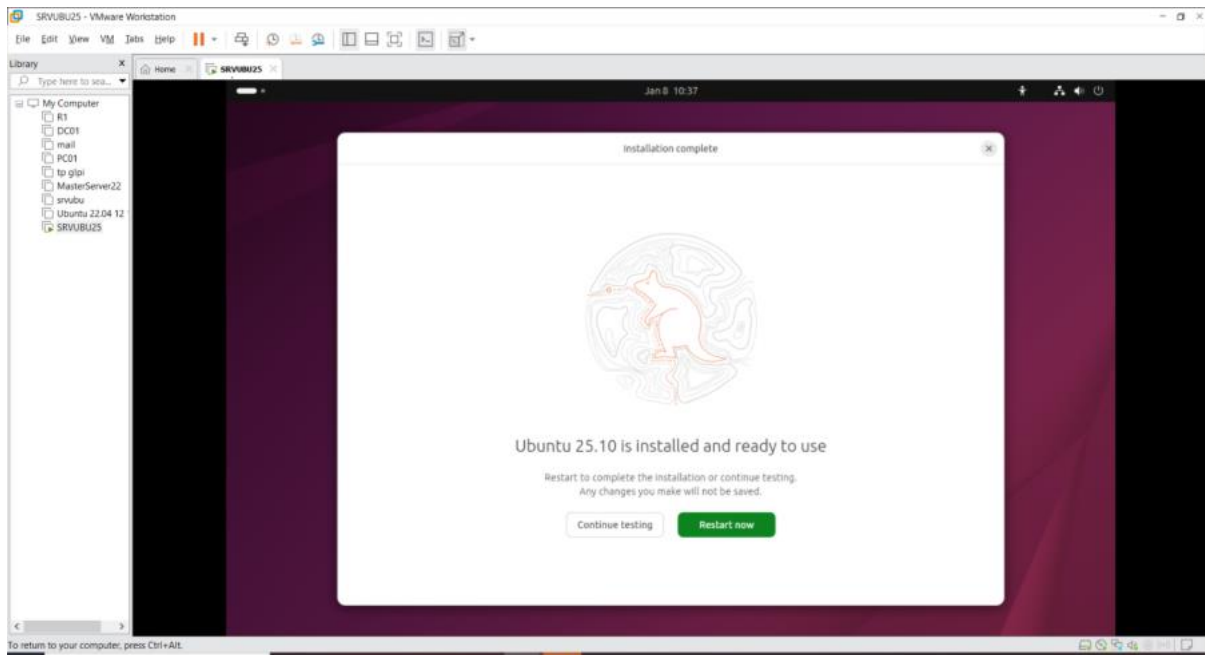












Modifiacation du fichier script.sh avec la cmd "nano script.sh" en renseignant ce texte : # Mettre à jour la liste des paquets disponibles à partir des sources configurées
apt update -y

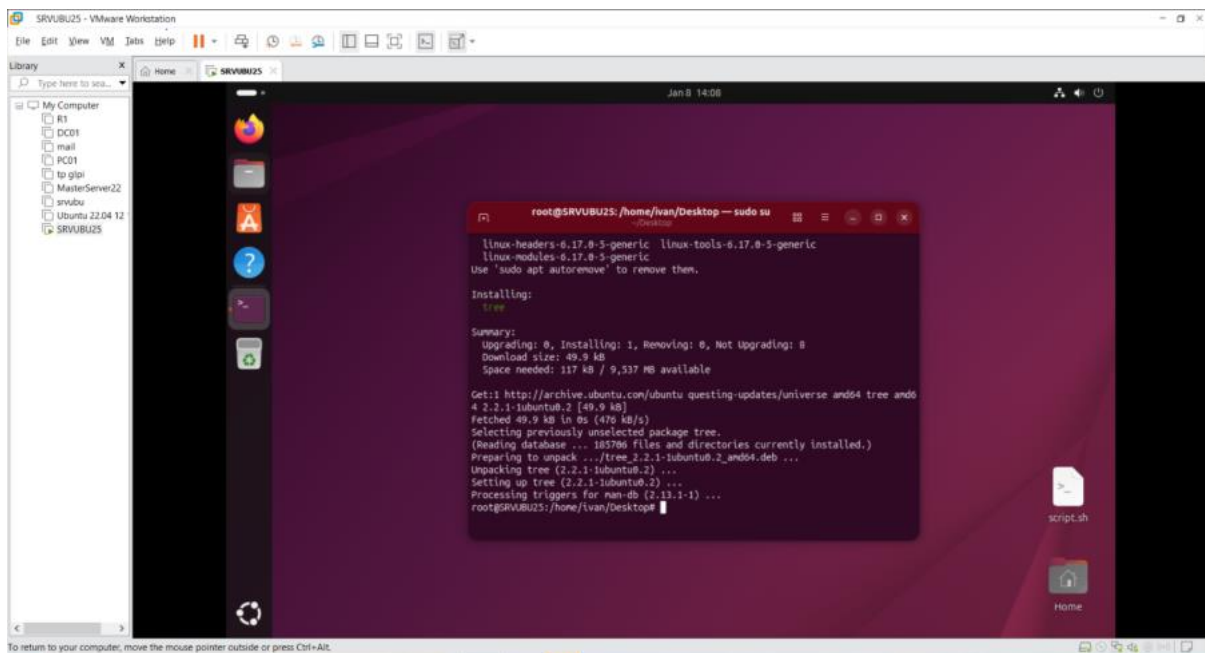
Effectuer une mise à niveau de distribution, incluant des mises à jour de paquets et des changements de dépendances
apt dist-upgrade -y

Installer le programme "tasksel" qui permet d'installer des tâches prédéfinies pour une configuration plus facile
apt install tasksel -y

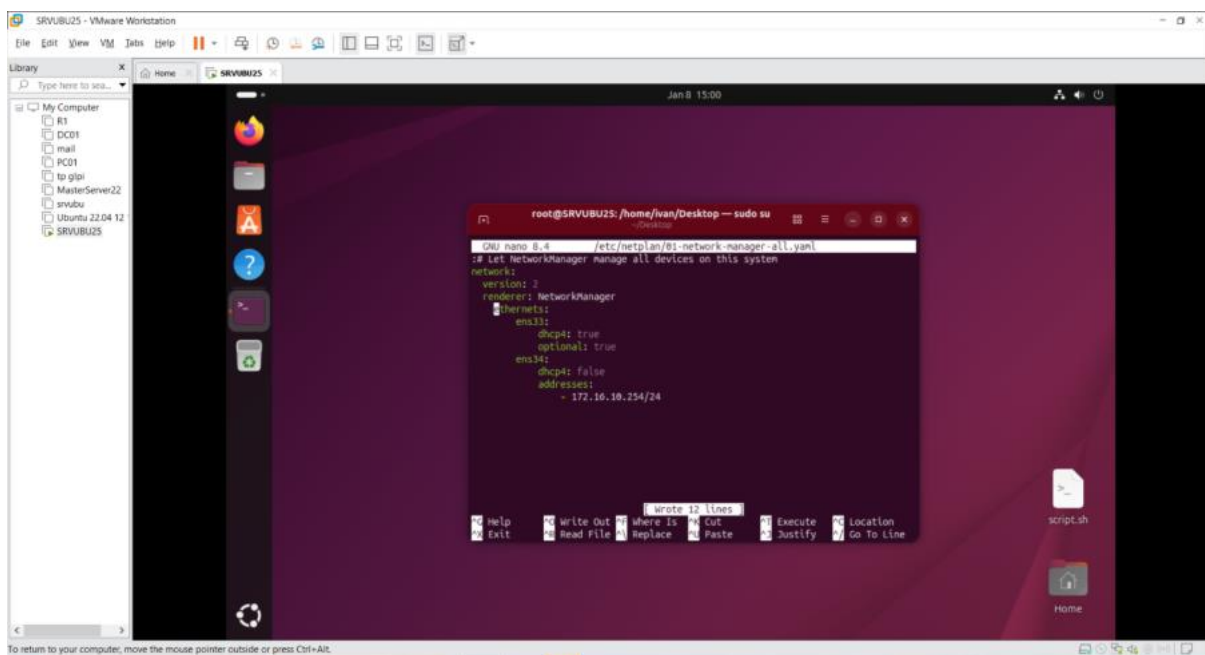
```
# Installer le serveur SSH, permettant l'accès à distance sécurisé via SSH
apt install openssh-server -y
```

```
# Installer les outils Open VM Tools, nécessaires pour les machines virtuelles
VMware
apt install open-vm-tools -y
```

```
# Installer le programme "tree" pour afficher les répertoires sous forme
d'arborescence
apt install tree -y
```

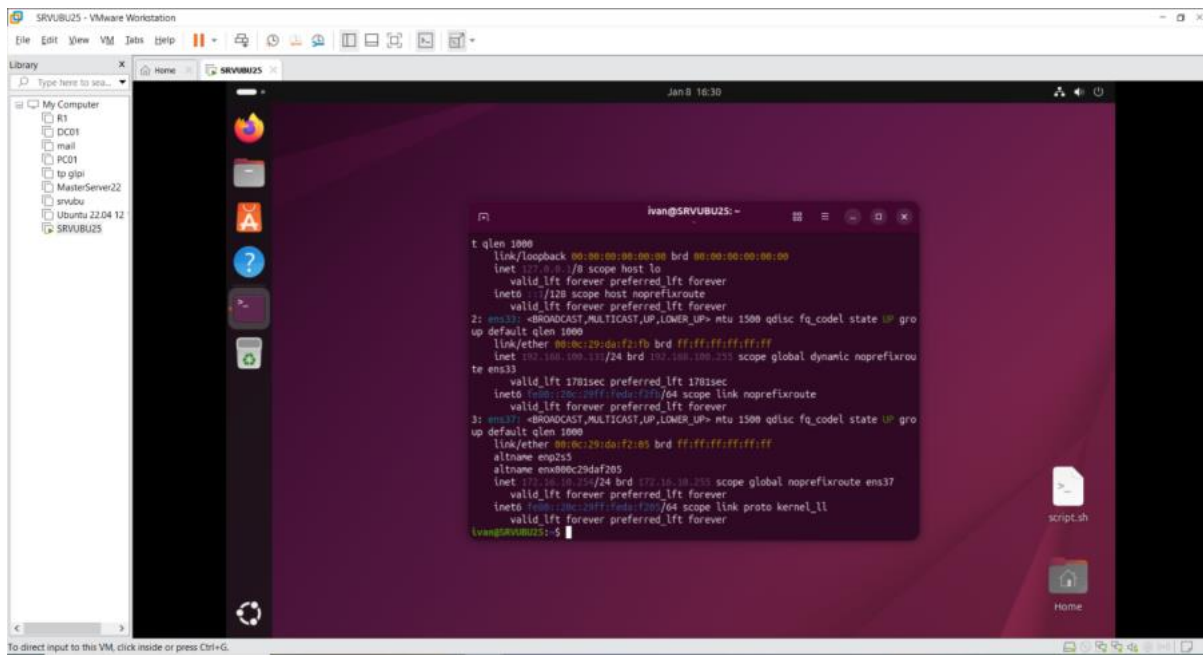


exécution de ce script avec la commande “./script.sh”



/etc/netplan/01-network-manager-all.yaml avec la commande “sudo nano


```
/etc/netplan/01-network-manager-all.yaml" et insertion du script : network:  
version: 2  
renderer: NetworkManager  
ethernets:  
  ens33:  
    dhcp4: true  
    optional: true  
  ens34:  
    dhcp4: false  
    addresses:  
      - 172.16.10.254/24
```



```
root@SRVUBU25: /home/ivan — sudo su
Get:1 http://archive.ubuntu.com/ubuntu questing/universe amd64 isc-dhcp-server a
md64 4.4.3-P1-4ubuntu2 [1,236 kB]
Get:2 http://archive.ubuntu.com/ubuntu questing/universe amd64 isc-dhcp-common a
md64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1,281 kB in 2s (759 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 185713 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Generating /etc/default/isc-dhcp-server...
Created symlink '/etc/systemd/system/multi-user.target.wants/isc-dhcp-server.ser
vice' → '/usr/lib/systemd/system/isc-dhcp-server.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.se
rvice' → '/usr/lib/systemd/system/isc-dhcp-server6.service'.
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Processing triggers for man-db (2.13.1-1) ...
root@SRVUBU25:/home/ivan#
```

adresses IP aux clients
du réseau avec la commande "sudo apt install isc-dhcp-server -y"

```
root@SRVUBU25: /home/ivan — sudo su
GNU nano 8.4 /etc/dhcp/dhcpd.conf
#   range 10.17.224.10 10.17.224.250;
# }
# pool {
#   deny members of "foo";
#   range 10.0.29.10 10.0.29.230;
# }
#}
#subnet 172.16.10.0 netmask 255.255.255.0 {
#   deny client-updates;
#   one-lease-per-client on;
#   option routers 172.16.10.254;
#   option broadcast-address 172.16.10.255;
#
#   pool {
#       deny dynamic bootp clients;
#       range 172.16.10.3 172.16.10.9;
#       host club22bis { hardware ethernet 00:0C:29:67:C6:8C; fixed-address 172.16.10.3;
#   }
#}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

de configuration DHCP avec la cmd "sudo nano
/etc/dhcp/dhcpd.conf"

```
root@SRVUBU25: /home/ivan — sudo su
GNU nano 8.4 /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens37"
#INTERFACESv6=""

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

INTERFACESv4 et #INTERFACESv6="", puis reboot du service DHCP ISC pour appliquer les nouvelles configurations avec la commande "sudo systemctl restart isc-dhcp-server"

4) Test du bon fonctionnement du DHCP

Pour tester le bon fonctionnement du dhcp il va falloir ping les machine club25,clwin10 et SRVUBU25 entres elles.

A) Machine club25 :


```
root@club25: /home/ivan
root@club25:/home/ivan# ping 172.16.10.10
PING 172.16.10.10 (172.16.10.10) 56(84) bytes of data.
64 bytes from 172.16.10.10: icmp_seq=1 ttl=64 time=0.038 ms
64 bytes from 172.16.10.10: icmp_seq=2 ttl=64 time=0.079 ms
64 bytes from 172.16.10.10: icmp_seq=3 ttl=64 time=0.092 ms
64 bytes from 172.16.10.10: icmp_seq=4 ttl=64 time=0.039 ms
64 bytes from 172.16.10.10: icmp_seq=5 ttl=64 time=0.077 ms
64 bytes from 172.16.10.10: icmp_seq=6 ttl=64 time=0.136 ms
64 bytes from 172.16.10.10: icmp_seq=7 ttl=64 time=0.078 ms
64 bytes from 172.16.10.10: icmp_seq=8 ttl=64 time=0.095 ms
64 bytes from 172.16.10.10: icmp_seq=9 ttl=64 time=0.095 ms
64 bytes from 172.16.10.10: icmp_seq=10 ttl=64 time=0.040 ms
64 bytes from 172.16.10.10: icmp_seq=11 ttl=64 time=0.116 ms
64 bytes from 172.16.10.10: icmp_seq=12 ttl=64 time=0.068 ms
^C
--- 172.16.10.10 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11242ms
rtt min/avg/max/mdev = 0.038/0.079/0.136/0.029 ms
root@club25:/home/ivan#
```

Ping de l'adresse 172.16.10.10 (la machine elle meme)

```
root@club25: /home/ivan
64 bytes from 172.16.10.10: icmp_seq=10 ttl=64 time=0.040 ms
64 bytes from 172.16.10.10: icmp_seq=11 ttl=64 time=0.116 ms
64 bytes from 172.16.10.10: icmp_seq=12 ttl=64 time=0.068 ms
^C
--- 172.16.10.10 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11242ms
rtt min/avg/max/mdev = 0.038/0.079/0.136/0.029 ms
root@club25:/home/ivan# ping 172.16.10.254
PING 172.16.10.254 (172.16.10.254) 56(84) bytes of data.
64 bytes from 172.16.10.254: icmp_seq=1 ttl=64 time=0.803 ms
64 bytes from 172.16.10.254: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 172.16.10.254: icmp_seq=3 ttl=64 time=5.92 ms
64 bytes from 172.16.10.254: icmp_seq=4 ttl=64 time=1.18 ms
64 bytes from 172.16.10.254: icmp_seq=5 ttl=64 time=1.28 ms
64 bytes from 172.16.10.254: icmp_seq=6 ttl=64 time=1.56 ms
64 bytes from 172.16.10.254: icmp_seq=7 ttl=64 time=1.44 ms
64 bytes from 172.16.10.254: icmp_seq=8 ttl=64 time=1.44 ms
64 bytes from 172.16.10.254: icmp_seq=9 ttl=64 time=1.48 ms
64 bytes from 172.16.10.254: icmp_seq=10 ttl=64 time=1.45 ms
^C
--- 172.16.10.254 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9019ms
rtt min/avg/max/mdev = 0.803/1.789/5.916/1.390 ms
root@club25:/home/ivan#
```

Ping de l'adresse 172.16.10.254 (SRVUBU25)

```
root@club25: /home/ivan
64 bytes from 172.16.10.254: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 172.16.10.254: icmp_seq=3 ttl=64 time=5.92 ms
64 bytes from 172.16.10.254: icmp_seq=4 ttl=64 time=1.18 ms
64 bytes from 172.16.10.254: icmp_seq=5 ttl=64 time=1.28 ms
64 bytes from 172.16.10.254: icmp_seq=6 ttl=64 time=1.56 ms
64 bytes from 172.16.10.254: icmp_seq=7 ttl=64 time=1.44 ms
64 bytes from 172.16.10.254: icmp_seq=8 ttl=64 time=1.44 ms
64 bytes from 172.16.10.254: icmp_seq=9 ttl=64 time=1.48 ms
64 bytes from 172.16.10.254: icmp_seq=10 ttl=64 time=1.45 ms
^C
--- 172.16.10.254 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9019ms
rtt min/avg/max/mdev = 0.803/1.789/5.916/1.390 ms
root@club25: /home/ivan# ping 172.16.10.03
PING 172.16.10.03 (172.16.10.3) 56(84) bytes of data.
64 bytes from 172.16.10.3: icmp_seq=1 ttl=128 time=0.841 ms
64 bytes from 172.16.10.3: icmp_seq=2 ttl=128 time=1.39 ms
64 bytes from 172.16.10.3: icmp_seq=3 ttl=128 time=1.01 ms
64 bytes from 172.16.10.3: icmp_seq=4 ttl=128 time=1.40 ms
^C
--- 172.16.10.03 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 0.841/1.160/1.399/0.242 ms
root@club25: /home/ivan#
```

Ping de l'adresse 172.16.10.03 (clwin10)

B) Machine clwin10 :

```
Invite de commandes
Microsoft Windows [version 10.0.10240]
(c) Microsoft Corporation, 2015. Tous droits réservés.

C:\Users\ivan>ping 172.16.10.03

Envoi d'une requête 'Ping' 172.16.10.3 avec 32 octets de données :
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128

Statistiques Ping pour 172.16.10.3:
    Paquets : envoyés = 3, recus = 3, perdus = 0 (perte 0%),
    Durée approximative des boucles en millisecondes :
        Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
Ctrl+C
^C
C:\Users\ivan>
```

Ping de l'adresse 172.16.10.03 (la machine elle meme)

```
Invite de commandes
Microsoft Windows [version 10.0.10240]
(c) Microsoft Corporation, 2015. Tous droits réservés.

C:\Users\ivan>ping 172.16.10.03

Envoi d'une requête 'Ping' 172.16.10.3 avec 32 octets de données :
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128
Réponse de 172.16.10.3 : octets=32 temps<1ms TTL=128

Statistiques Ping pour 172.16.10.3:
    Paquets : envoyés = 3, recus = 3, perdus = 0 (perte 0%),
    Durée approximative des boucles en millisecondes :
        Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
Ctrl+C
^C
C:\Users\ivan>ping 172.16.10.10

Envoi d'une requête 'Ping' 172.16.10.10 avec 32 octets de données :
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64

Statistiques Ping pour 172.16.10.10:
    Paquets : envoyés = 4, recus = 4, perdus = 0 (perte 0%),
    Durée approximative des boucles en millisecondes :
        Minimum = 0ms, Maximum = 6ms, Moyenne = 1ms
C:\Users\ivan>
```

Ping de l'adresse 172.16.10.10 (club25)

```
Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
Ctrl+C
^C
C:\Users\ivan>ping 172.16.10.10

Envoi d'une requête 'Ping' 172.16.10.10 avec 32 octets de données :
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps=6 ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.10 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 172.16.10.10:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 6ms, Moyenne = 1ms

C:\Users\ivan>ping 172.16.10.254

Envoi d'une requête 'Ping' 172.16.10.254 avec 32 octets de données :
Réponse de 172.16.10.254 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.254 : octets=32 temps<1ms TTL=64
Réponse de 172.16.10.254 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 172.16.10.254:
    Paquets : envoyés = 3, reçus = 3, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 1ms, Moyenne = 0ms
Ctrl+C
^C
C:\Users\ivan>
```

Ping de l'adresse 172.16.10.254 (SRVUBU25)

C) Machine SRVUBU25 :

```
root@SRVUBU25: /etc/bind — sudo su
root@SRVUBU25: /etc/bind# ping 172.16.10.254
PING 172.16.10.254 (172.16.10.254) 56(84) bytes of data.
64 bytes from 172.16.10.254: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 172.16.10.254: icmp_seq=2 ttl=64 time=0.101 ms
^C
--- 172.16.10.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.043/0.072/0.101/0.029 ms
root@SRVUBU25: /etc/bind#
```



```
root@SRVUBU25: /etc/bind — sudo su
root@SRVUBU25:/etc/bind# ping 172.16.10.254
PING 172.16.10.254 (172.16.10.254) 56(84) bytes of data.
64 bytes from 172.16.10.254: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 172.16.10.254: icmp_seq=2 ttl=64 time=0.101 ms
^C
--- 172.16.10.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.043/0.072/0.101/0.029 ms
root@SRVUBU25:/etc/bind# ping 172.16.10.10
PING 172.16.10.10 (172.16.10.10) 56(84) bytes of data.
64 bytes from 172.16.10.10: icmp_seq=1 ttl=64 time=0.505 ms
64 bytes from 172.16.10.10: icmp_seq=2 ttl=64 time=1.26 ms
64 bytes from 172.16.10.10: icmp_seq=3 ttl=64 time=1.39 ms
64 bytes from 172.16.10.10: icmp_seq=4 ttl=64 time=1.09 ms
^C
--- 172.16.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 0.505/1.061/1.392/0.338 ms
root@SRVUBU25:/etc/bind#
```

```
root@SRVUBU25: /etc/bind — sudo su
64 bytes from 172.16.10.254: icmp_seq=2 ttl=64 time=0.101 ms
^C
--- 172.16.10.254 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.043/0.072/0.101/0.029 ms
root@SRVUBU25:/etc/bind# ping 172.16.10.10
PING 172.16.10.10 (172.16.10.10) 56(84) bytes of data.
64 bytes from 172.16.10.10: icmp_seq=1 ttl=64 time=0.505 ms
64 bytes from 172.16.10.10: icmp_seq=2 ttl=64 time=1.26 ms
64 bytes from 172.16.10.10: icmp_seq=3 ttl=64 time=1.39 ms
64 bytes from 172.16.10.10: icmp_seq=4 ttl=64 time=1.09 ms
^C
--- 172.16.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 0.505/1.061/1.392/0.338 ms
root@SRVUBU25:/etc/bind# ping 172.16.10.03
PING 172.16.10.03 (172.16.10.3) 56(84) bytes of data.
64 bytes from 172.16.10.3: icmp_seq=1 ttl=128 time=0.703 ms
64 bytes from 172.16.10.3: icmp_seq=2 ttl=128 time=1.47 ms
^C
--- 172.16.10.03 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.703/1.088/1.474/0.385 ms
root@SRVUBU25:/etc/bind#
```